

realman Product Selection Manual

MTBF 50000H
Hours

ROHS/Reach
Compliance Certified

RealMan Group

Sales@realman-robot.com
<https://www.realman-robotics.com>



RealMan Group is a system-level infrastructure platform company built for the era of embodied intelligence.

RealMan Group is a system-level infrastructure platform company positioned for the era of embodied intelligence.

With “hardware + data + Global Link Network (GLN)” as its three foundational capability flywheels, the Group establishes an integrated development framework spanning from actuation joints to robotic manipulators and complete robotic systems, from cross-temporal and cross-spatial labor capability reconstruction networks to data assets, and from global robot superfactories to an open ecosystem architecture.

Through continuous technological iteration and capability scaling of embodied intelligence, RealMan Group accelerates the realization of its vision of deploying robotics to serve human society.

Products & Services

As a system-level infrastructure platform company for the era of embodied intelligence, RealMan has, since its founding in 2018, continuously driven technological innovation in embodied-motion robotic arms and precision, high-durability joint modules. Its self-developed integrated joint modules now exceed an annual production capacity of 100,000 units. The robotic arms have obtained CR L3 certification, with an average MTBF of 50,000 hours, providing a solid foundation for stable and continuous robot operation. RealMan’s business footprint spans Asia, Europe, North America, and South America, serving over 8,000 enterprise customers worldwide through partnerships with dozens of leading global enterprises and institutions.

MTBF 50000H
Hours

ROHS/Reach
Compliance Certified



06/34
Power Joint
Series

14/34
Humanoid Robotic Arm
Series

24/34
ODM Customized
Arms

28/34
High-Performance
Robots

34/34
Embodied Data
Services



WHJ Series

Precision in Compact Form Precise Control at Every Degree

All-scenario compatibility:
supports various robotic arms, robotic systems, grippers,
precision rotary tables, and specialized tools.



High Torque
Density



High
Reliability



High
Precision



WHJ10-B



WHJ10-N

Model	WHJ10-B	WHJ10-N
Unit Weight (g)	455	445
Dimensions (Diameter × Length mm)	50*73	50*64
Hollow Shaft Diameter (mm)	8	8
Rated Voltage (V)	24/48	24/48
Rated Current (A)	1.3A (48V)	1.3A (48V)
Rated Power (W)	63	63
Peak Speed (RPM)	75 (48V)	75 (48V)
Rated Torque (N·m)	10	10
Gear Ratio	80	80
Operating Temperature (°C)	0 ~ 50	0 ~ 50
Incremental Encoder (bits)	16	16
Absolute Encoder (bits)	18	18
Brake Type	Pin-Type Brake	Spring-Applied Brake
Communication Interface	CANFD	CANFD



Model	WHJ03	WHJ30
Unit Weight (g)	170	750
Dimensions (Diameter × Length mm)	33*48	60*86.5
Hollow Shaft Diameter (mm)	4	9
Rated Voltage (V)	24/48	24/48
Rated Current (A)	0.4A (48V)	2.8A (48V)
Rated Power (W)	20	137
Peak Speed (RPM)	60 (48V)	75 (48V)
Rated Torque (N·m)	3	30
Gear Ratio	100	80
Operating Temperature (°C)	0 ~ 50	0 ~ 50
Incremental Encoder (bits)	16	16
Absolute Encoder (bits)	18	18
Brake Type	Spring-Applied Brake	Pin-Type Brake
Communication Interface	CANFD	CANFD



Model	WHJ60	WHJ120
Unit Weight (g)	1040	2300
Dimensions (Diameter × Length mm)	70*91	88*109
Hollow Shaft Diameter (mm)	9	16
Rated Voltage (V)	24/48	24/48
Rated Current (A)	5.3A (48V)	17.8
Rated Power (W)	264	857
Peak Speed (RPM)	60 (48V)	30
Rated Torque (N·m)	60	120
Gear Ratio	100	100
Operating Temperature (°C)	0 ~ 50	0 ~ 50
Incremental Encoder (bits)	16	16
Absolute Encoder (bits)	18	18
Brake Type	Pin-Type Brake	Electromagnetic Brake
Communication Interface	CANFD	CANFD



WHJ Torque Series

Joint-Level Force Sensing Master Every Interaction

Comprehensive Force Control Solutions: Precision assembly, grinding & polishing, collaborative collision detection, flexible grasping, and cross-platform robotic compatibility.



High-Precision
Force Sensing



Dynamic Force
Closed-Loop Control



Compliant
Interaction



Model	WHJ10 Torque Version	WHJ30 Torque Version	WHJ60 Torque Version
Unit Weight (g)	489	820	1121
Dimensions (Diameter × Length mm)	50*78	60*101	70*106
Hollow Shaft Diameter (mm)	8	7	9
Rated Voltage (V)	24	24	24
Rated Current (A)	4	11	14
Rated Power (W)	90	256	336
Peak Speed (RPM)	37.5	37.5	30
Rated Torque (N·m)	10	30	60
Gear Ratio	80	80	100
Operating Temperature (°C)	0 ~ 50	0 ~ 50	0 ~ 50
Incremental Encoder (bits)	16	16	16
Absolute Encoder (bits)	18	18	18
Brake Type	Spring-Applied Brake	Pin-Type Brake	Pin-Type Brake
Communication Interface	CANFD	CANFD	CANFD



WHG Series

High-Performance Robotic Power Modules

Suitable for industrial automation, collaborative robots, humanoid robots, semiconductors, and precision equipment applications



High Power
Density



High
Rigidity



High
Precision



WHG1410



WHG1730



WHG2060



WHG25120



WHG32240

Model	WHG1410	WHG1730	WHG2060	WHG25120	WHG32240
Unit Weight (g)	800	1308	1770	2816	5773
Dimensions (Diameter × Length mm)	70*69	80*80	90*92	110*103	142*127
Hollow Shaft Diameter (mm)	9	9	9	16	18
Rated Voltage (V)	24/48	24/48	24/48	24/48	24/48
Rated Current (A)	1.5	3	6	13	18
Rated Power (W)	73	155	287	611	873
Peak Speed (RPM)	75	75	60	45	27
Rated Torque (N·m)	10	30	60	120	240
Gear Ratio	100	100	100	100	100
Operating Temperature (°C)	0 ~ 50	0 ~ 50	0 ~ 50	0 ~ 50	0 ~ 50
Incremental Encoder (bits)	16	16	16	16	16
Absolute Encoder (bits)	18	18	18	18	18
Brake Type	Electromagnetic Brake	Electromagnetic Brake	Electromagnetic Brake	Electromagnetic Brake	Electromagnetic Brake
Communication Interface	CANFD	CANFD	CANFD	CANFD	CANFD



RM Series

Lightweight Yet Powerful, Handles Complex Scenarios with Ease

Suitable for new retail, food & beverage, lifestyle services,
healthcare, and exhibition applications`



High
Payload



Force
Sensing



Integrated
Design



RM65
Standard Version



RM65
Six-Axis Force Version



RM75
Standard Version



RM75
Six-Axis Force Version

Model	RM65 Standard Version	RM65 Six-Axis Force Version	RM75 Standard Version	RM75 Six-Axis Force Version
Degrees of Freedom	6	6	7	7
Configuration	Humanoid Configuration	Humanoid Configuration	Humanoid Configuration	Humanoid Configuration
Joint Brake Type	Mechanical Brakes on Joints 1-3 Spring-Applied Brakes on Joints 4-6	Mechanical Brakes on Joints 1-3 Spring-Applied Brakes on Joints 4-6	Mechanical Brakes on Joints 1-4 Spring-Applied Brakes on Joints 5-7	Mechanical Brakes on Joints 1-4 Spring-Applied Brakes on Joints 5-7
Working Radius (mm)	610	627	610	627
Payload (kg)	5	5	5	5
Net Weight (kg)	7.2	7.3	7.8	7.9
Repeatability (mm)	±0.05	±0.05	±0.05	±0.05
TCP Linear Speed (m/s)	≤1.8	≤1.8	≤1.8	≤1.8
Typical Power (W)	≤150	≤150	≤200	≤200
Peak Power (W)	≤900	≤900	≤1000	≤1000
Base Diameter (mm)	φ107	φ107	φ107	φ107
Operating Temperature (°C)	0-45	0-45	0-45	0-45
Six-Axis Force Range	—	200N/7N·m	—	200N/7N·m
Six-Axis Force Accuracy	—	±0.5%FS	—	±0.5%FS
Power Supply (V)	DC 24V	DC 24V	DC 24V	DC 24V



RML Series

Extended Reach, Beyond the Standard Work Radius

Suitable for new retail, food & beverage, lifestyle services, healthcare, and exhibition applications



Large Working
Radius



Force
Sensing



Application-
Grade



RML63
Standard Version



RML63
Six-Axis Force Version

Model	RML63 Standard Version	RML63 Six-Axis Force Version
Degrees of Freedom	6	6
Configuration	Humanoid Configuration	Humanoid Configuration
Joint Brake Type	Mechanical Brakes on Joints 1-4 Spring-Applied Brakes on Joints 5-6	Mechanical Brakes on Joints 1-4 Spring-Applied Brakes on Joints 5-6
Working Radius (mm)	900	917
Payload (kg)	3	3
Net Weight (kg)	10	10.1
Repeatability (mm)	±0.05	±0.05
TCP Linear Speed (m/s)	≤2.8	≤2.8
Typical Power (W)	≤150	≤150
Peak Power (W)	≤900	≤900
Base Diameter (mm)	φ107	φ107
Operating Temperature (°C)	0-45	0-45
Six-Axis Force Range	—	200N/7N·m
Six-Axis Force Accuracy	—	±0.5%FS
Power Supply (V)	DC 24V	DC 24V



High Cost-Performance Choice, Precise, Efficient Basics

Suitable for new retail, industrial production lines, massage therapy, coffee robotics, and medical applications



Compact-
Class



Economical
Model



Low Energy
Consumption

		
Model	ECO65 Standard Version	ECO65 Six-Axis Force Version
Degrees of Freedom	6	6
Configuration	Collaborative Robot Configuration	Collaborative Robot Configuration
Joint Brake Type	Mechanical Brakes on Joints 1-6	Mechanical Brakes on Joints 1-6
Working Radius (mm)	610	627
Payload (kg)	5	5
Net Weight (kg)	7.8	7.9
Repeatability (mm)	±0.05	±0.05
TCP Linear Speed (m/s)	≤1.8	≤1.8
Typical Power (W)	≤150	≤150
Peak Power (W)	≤900	≤900
Base Diameter (mm)	φ107	φ107
Operating Temperature (°C)	0-45	0-45
Six-Axis Force Range	—	200N/7N·m
Six-Axis Force Accuracy	—	±0.5%FS
Power Supply (V)	DC 24V	DC 24V



ECO63
Standard Version



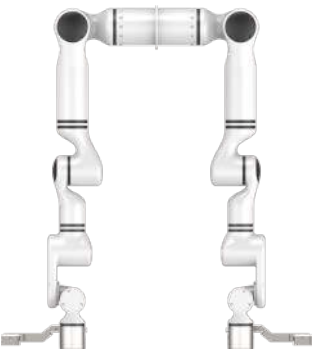
ECO63
Six-Axis Force Version

Model		
Degrees of Freedom	6	
Configuration	Collaborative Robot Configuration	
Joint Brake Type	Mechanical Brakes on Joints 1–6	
Working Radius (mm)	900	917
Payload (kg)	3	3
Net Weight (kg)	9.5	9.6
Repeatability (mm)	±0.05	±0.05
TCP Linear Speed (m/s)	≤2.8	≤2.8
Typical Power (W)	≤150	≤150
Peak Power (W)	≤900	≤900
Base Diameter (mm)	φ107	φ107
Operating Temperature (°C)	0–45	0–45
Six-Axis Force Range	—	200N/7N·m
Six-Axis Force Accuracy	—	±0.5%FS
Power Supply (V)	DC 24V	DC 24V



ECO62
Standard Version

Model		
Degrees of Freedom	6	
Configuration	Collaborative Robot Configuration	
Joint Brake Type	Mechanical Brakes on Joints 1–6	
Working Radius (mm)	355	
Payload (kg)	1	
Net Weight (kg)	3.3	
Repeatability (mm)	±0.1mm	
TCP Linear Speed (m/s)	< 1.3	
Typical Power (W)	≤100	
Peak Power (W)	≤250	
Base Diameter (mm)	φ107	
Operating Temperature (°C)	0–50	
Six-Axis Force Range	—	
Six-Axis Force Accuracy	—	
Power Supply (V)	DC 24V	



Model

RX75
Standard Version

RX75
Vision Version

Degrees of Freedom

7

7

Configuration

Humanoid Configuration

Humanoid Configuration

Joint Brake Type

Mechanical Brakes on Joints 1-4
Spring-Applied Brakes on Joints 5-7

Mechanical Brakes on Joints 1-4
Spring-Applied Brakes on Joints 5-7

Working Radius (mm)

709

732

Payload (kg)

5

5

Net Weight (kg)

8.2

8.4

Repeatability (mm)

±0.2

±0.2

TCP Linear Speed (m/s)

≤2.2

≤2.2

Typical Power (W)

≤200

≤200

Power Supply (V)

DC20~30V

DC20~30V

End-Effector Camera

—

D405

Operating Temperature (°C)

0-45

0-45

Six-Axis Force Range

200N/7N·m

200N/7N·m

Six-Axis Force Accuracy

0.5%FS

0.5%FS

Material

Aluminum Alloy / ABS

Aluminum Alloy / ABS

RX Series

Humanoid Wrist Design, Mirrors Human Arm Motion

Suitable for side-mounted dual-arm collaborative operations, widely applicable in retail, food service, industry, warehousing, and medical humanoid applications



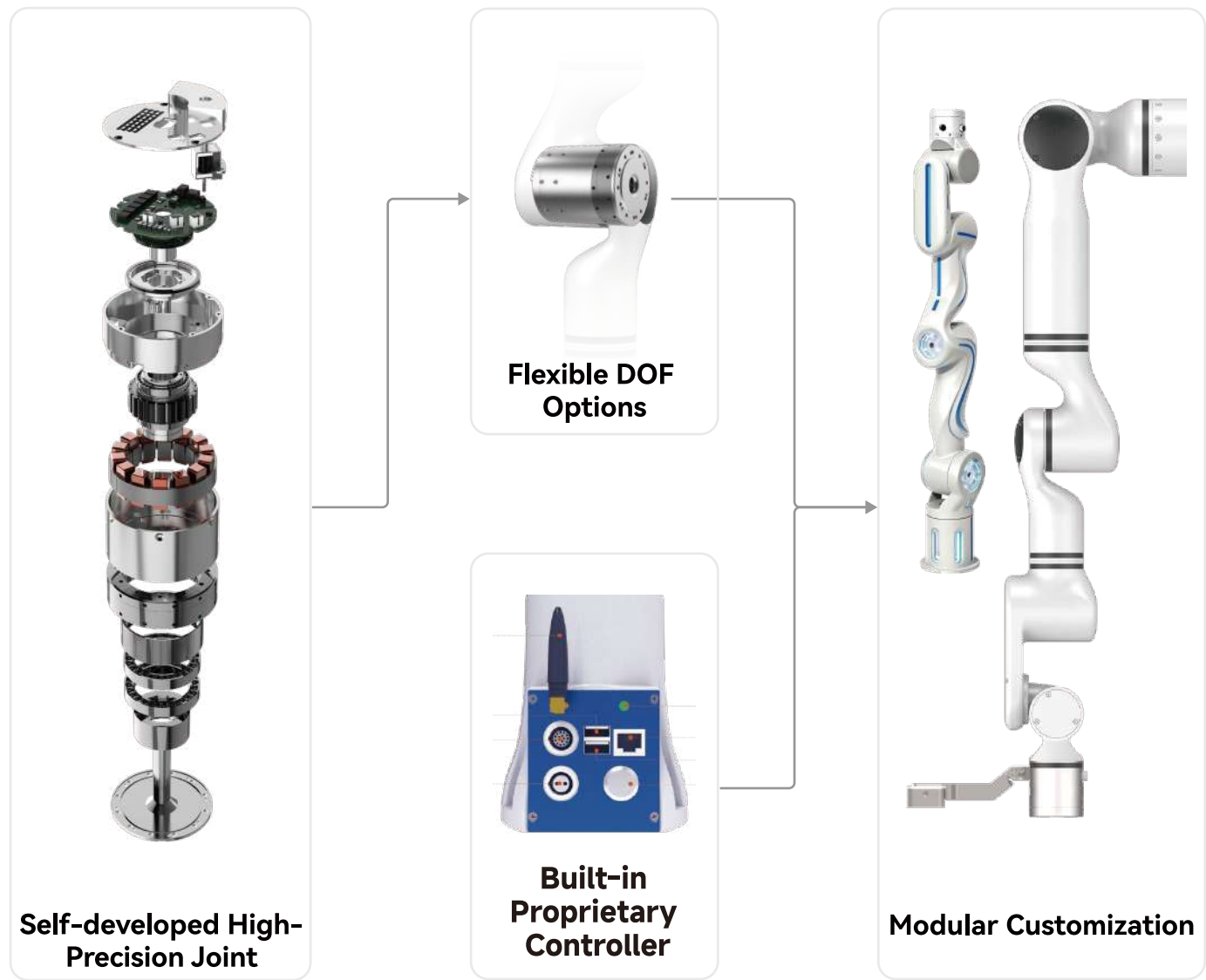
Lightweight
Design



Human
oid Form



Integrated
Design



Robotic Arm ODM

RealMan provides comprehensive customized robotic arm solutions. Based on self-developed core joints and motion control technologies, it supports flexible configuration of 6-axis, 7-axis and other DOF structures to meet high-precision operation requirements in different scenarios. Whether industrial assembly, scientific research experiments, or human-robot collaboration applications, personalized designs can be quickly responded to.

Case Story

A leading domestic embodied robotics enterprise required a customized robotic arm with 7 degrees of freedom and 3kg rated payload for its self-developed model and chose to cooperate with our company. Relying on mature production lines, we achieved rapid batch delivery. Product accuracy and operational stability simultaneously met standards, efficiently matching the customer's mass production pace, and gained high praise for reliable quality and delivery efficiency.

A domestic emerging robotics manufacturer customized a dedicated robotic arm with 7 degrees of freedom and 5kg rated payload for service scenario deployment. Our company precisely matched its structural configuration and operational requirements, completed small-batch delivery, and ensured that arm flexibility and payload performance perfectly adapted to scenario operations, helping customers quickly advance product deployment testing.



Fast ODM
30-Day Delivery



Factory Capacity Nearly
1 Million Units



Serving Over
8,000+ Customers



Complete Robot ODM

RealMan robots adopt a modular architecture divided into four major modules: chassis, legs, torso, and arms. Standardized interfaces enable independent customization and rapid integration, balancing performance, stability, and scalability.

- Chassis: Supports multiple mobility configurations, adaptable to various scenarios;
- Torso: Integrated core for computing, perception, and interaction; supports flexible sensor-suite configuration.
- Arms: Built on proprietary joint modules, supporting multiple degrees of freedom and configurable specifications for precision tasks.
- Camera: Core vision sensor with GMSL interface as standard—low latency, strong anti-interference performance, and high reliability; industrial-grade stability with no frame drops.
- End-Effector:
Compatible with mainstream grippers and dexterous hands from leading vendors such as Zhixing, BrainCo , and OYMotion.

This design enables rapid response to differentiated requirements, shortens delivery cycles, and reduces customization and maintenance costs.



Fast ODM
30-Day Delivery



Factory Capacity Nearly
1 Million Units



Serving Over
8,000+ Customers



RealBOT Robot

Overall Dimensions:	570mm×465mm×1680mm
Turning Radius:	690mm (in-place rotation)
Working Range:	Max vertical reach: 2100mm Horizontal reach: 1818mm (excluding end effector)
Total Weight:	Approx. 125kg
Single Arm Payload:	5 kg
Total Active DOF:	21
Movement Speed:	0.1 m/s ~ 1.5 m/s
Battery & Endurance:	40AH (with BMS management system)
Charging Time:	~2 hours
Vision System:	Dual stereo depth / RGB optional
Wide-Angle Cameras:	130° field of view (3 units located on chest, back, and base)
LiDAR:	Single-line laser, detection range >40m (90% reflectivity)
Voice Interaction Module:	Circular 6-mic smart speaker (optional)

RealBOT is a comprehensive embodied intelligence development platform designed to accelerate innovation in embodied AI. It integrates advanced full-body motion control, multi-dimensional perception, and precise manipulation capabilities, providing an ideal hardware platform for researchers and developers.

Main Applications: AI training & data collection, new retail, education, industrial production lines, etc.



Dual-Arm Vertical Lift Robot

Overall Dimensions:	640x450x1780mm
Turning Radius:	450mm (in-place rotation)
Working Range:	Vertical: 0-2400mm (fully extended) Lateral: single side extends 790mm beyond chassis Forward: 0-690mm
Total Weight:	Approx. 157kg
Single Arm Payload:	5 kg
Effective Lifting Stroke:	900mm
Movement Speed:	1.5m/s
Endurance:	3~4h
Charging Power Supply:	AC 220V ±10% 50Hz
Robot Arm Model:	RM65-B-V (optional: RM75-B-V)
Vision Sensor:	Yes
End Effector:	CTAG2F90D (standard gripper)
Operating Temperature:	5~40°C

The platform supports teleoperation development, dual-arm data collection and model training, as well as autonomous inspection and task execution across various scenarios, covering industrial, commercial, household and research applications.

Main Applications: AI training & data collection, new retail, education, industrial production lines, etc.

Core Capabilities: Full-body motion control, multi-dimensional perception, precise manipulation.



Single-Arm Vertical Lift Robot

Overall Dimensions:	Total height 1605mm; Diameter 505mm
Charging Power Supply:	100~240V AC 50Hz
Total Weight:	Approx. 100kg
Single Arm Payload:	5 kg
Effective Lifting Stroke:	≥900mm
Movement Speed:	1m/s
Endurance:	3h
Power Supply:	DC24V
Robot Arm Model:	RM65-B-V
Positioning Accuracy:	< ±5cm
Turning Radius:	252.5mm
Operating Temperature:	5~40°C

The robot autonomously plans and executes inspection and operation tasks across diverse scenarios, enabling broad applications in industrial, commercial, home, and research settings.

Main Applications: New retail, logistics, etc.

Core Capabilities: Full-body motion control and precise manipulation.

60,000

Trajectories per Day

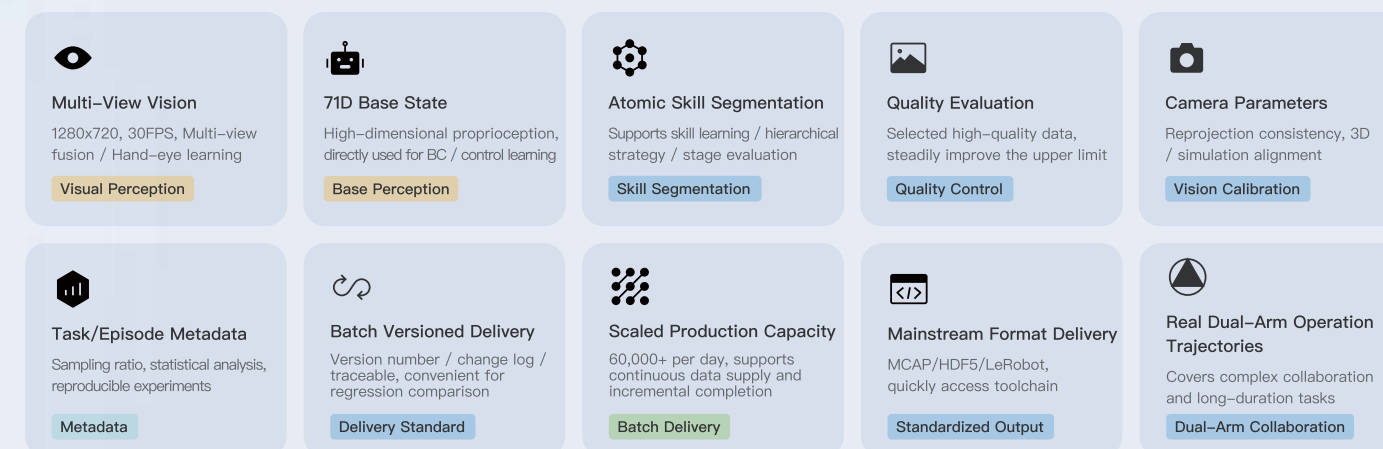
Scalable Delivery

Training-Grade
Alignment

Hardware-Level Synchronization

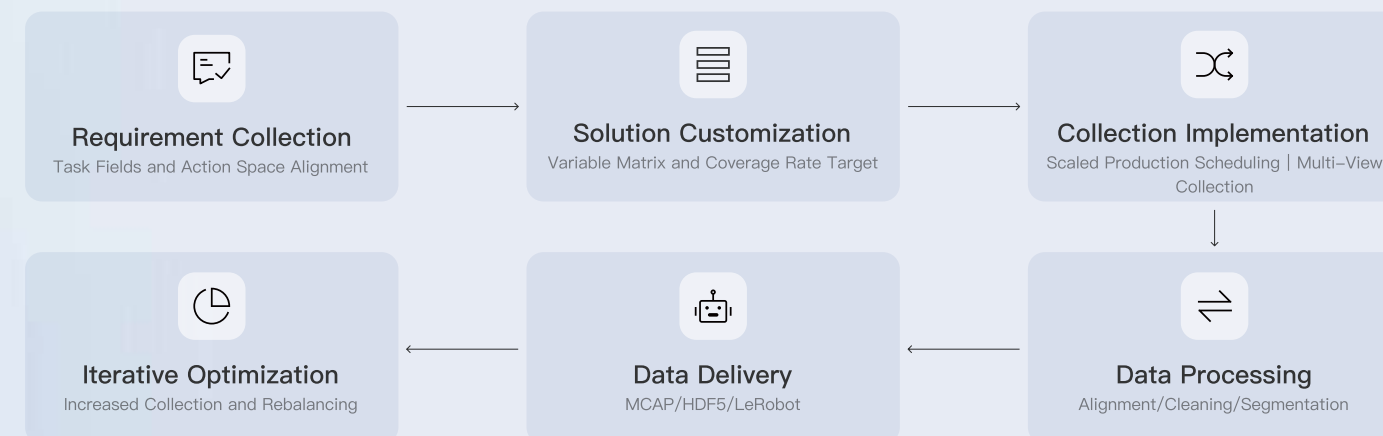
71D Base State
Complete Data ModalitiesReal-World Multimodal Trajectory
Data for Embodied Model Training

Delivery Components / Training Value



Value Proposition

We deliver **train-ready** real-world trajectories featuring multi-view observations with state-action closed loops, combined with atomic skill segmentation and quality evaluation. Supports mainstream formats for rapid integration into training and evaluation pipelines.



Scan to Access Open Dataset

Open Dataset (partial) : RealSource World